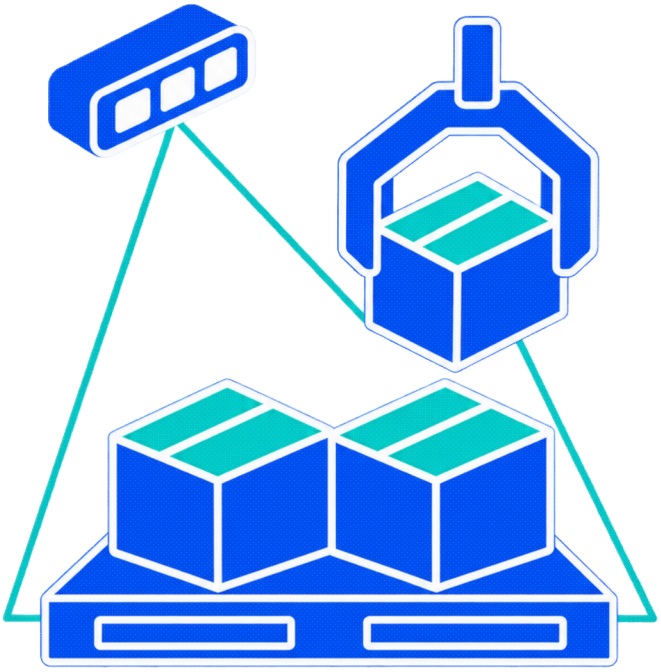
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AW3 PalletEye

Depalletizing Algorithm User Manual

Detect bags and boxes. Output pick positions, orientations and sequence.

Zhejiang MRDVS Technology Co., Ltd.

V1.0 · 2026-11-10

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1 Product Overview

PalletEye detects top-layer packages—bags or boxes—and outputs their pick positions, orientations and sequence.

AW3 manages device connections, camera groups, communication and logs. PalletEye handles calibration, package settings, recognition and localization.

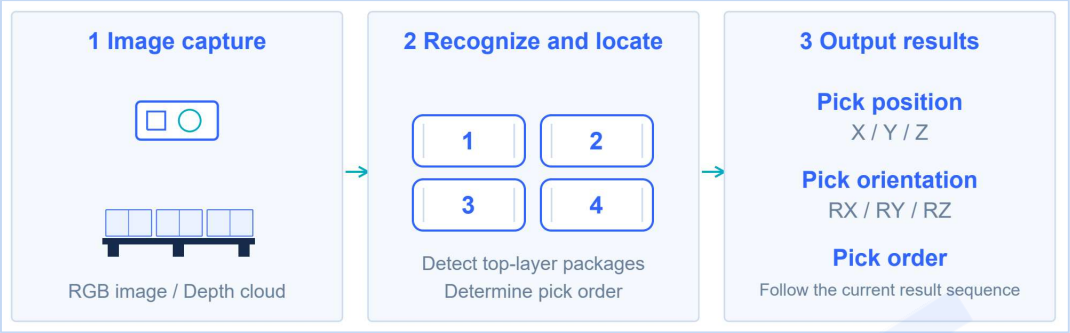


Figure 1. Depalletizing workflow (illustration)

1.1 Installation and Connections

The system consists of an M10 camera and a compute unit. The compute unit is the AW3 center and runs Ubuntu Linux and the depalletizing algorithm.

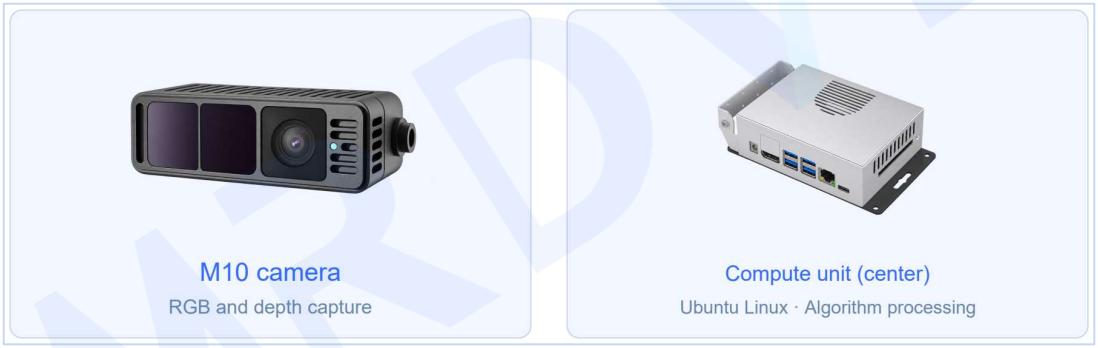


Figure 2. M10 camera and compute unit

Choose either operating method to suit the site. In both cases, the compute unit processes the camera data.

Operating method	Connections	Software
Operate on the compute unit	Connect a monitor to the compute unit. Connect the camera and compute unit to a network switch.	Use the software installed on the compute unit.
Operate from a Windows PC	Connect the camera, compute unit and PC to the same switch.	Install AW3 for Windows on an industrial PC or laptop.

Note: Product images identify the devices; they are not to scale. Provide a switch, monitor and PC as required.

For AW3 installation, device connections and camera settings, see the AW3 User Manual.

1.1.1 Operate on the Compute Unit

Connect a monitor directly to the compute unit and operate the software in its Ubuntu Linux environment.

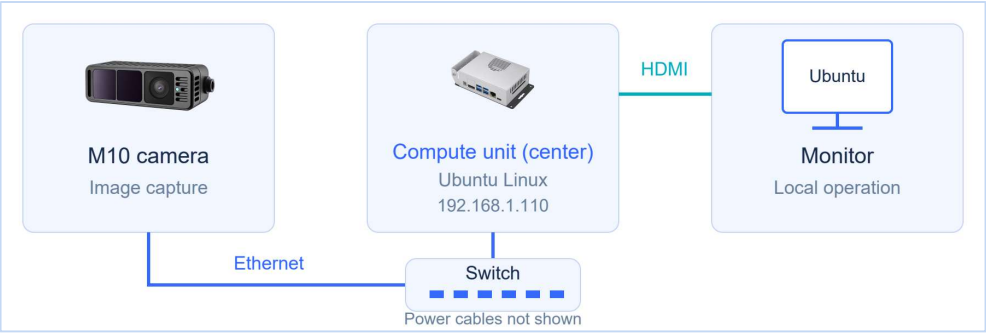


Figure 3. Method 1: local operation

1. Connect the M10 camera and compute unit to the same switch with Ethernet cables.
2. Connect a monitor to the compute unit with an HDMI cable, then connect a keyboard and mouse. Power each device as specified in its manual.
3. Sign in to Ubuntu Linux on the compute unit and open the installed software.
4. Check the IP settings described below. In Device Connection, confirm the center and M10 camera, then check images, point clouds and device status.

1.1.2 Operate from a Windows PC

Install AW3 on a Windows industrial PC or laptop, then connect to the compute unit over the network.

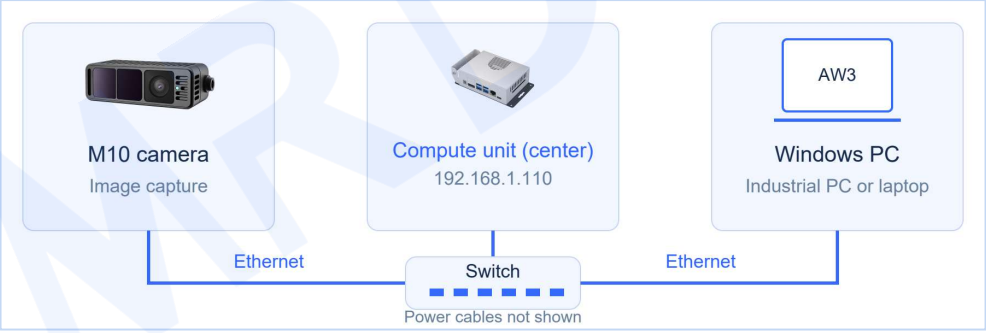


Figure 4. Method 2: Windows PC operation

1. Connect the M10 camera, compute unit and Windows PC to the same switch with Ethernet cables. Power on the devices.
2. Download and install AW3 for Windows. Clear the option to download the AW3 backend during installation.
3. Assign different IP addresses on the same subnet as described below. In AW3, open Device Connection and connect to the center at 192.168.1.110.
4. Select the M10 camera and camera group. Confirm that RGB images, depth clouds and device status are normal.

Note: AW3 runs on the Windows PC, while the algorithm runs on the compute unit. If its default IP address has changed, connect using the new address.

Network Settings (Both Methods)

The camera, compute unit and Windows PC must be on the same subnet. Each device must have a unique IP address.

Device	IP address	Description
Compute unit (center)	192.168.1.110	Default address used by AW3 to connect to the center.
M10 camera	192.168.1.100	Example only; use the camera's actual address.
Windows PC	192.168.1.200	Example only; required for Method 2.

Note: These examples use subnet mask 255.255.255.0. If the site uses another subnet, update and record all addresses to avoid IP conflicts.

Important: Before first use, confirm algorithm authorization and deploy the Soft Pack Depalletizing camera parameters in AW3. Confirm deployment before configuring stations, calibrating or testing. See Algorithm Authorization and Base Camera in the AW3 User Manual.

For AW3 installation, device connections and camera settings, see the AW3 User Manual.

1.2 First-Time Setup Workflow

Important: Before configuration, check the installation against Section 2.4. Mount the camera securely and level, parallel to the load-supporting surface, with its lens pointing straight down. Keep the entire stack visible from maximum height to the final layer.



Use Expert for configuration and Monitor to view results. Create separate material templates for bags and boxes.

After connecting the devices and deploying camera parameters, follow Chapter 3 in order. Use Single Trigger to check recognition and actual picking.

2 Before You Begin

2.1 Supported Packages and Stacking Patterns

Item	Conditions
Bags	Some deformation, close contact and underfilling are acceptable. Bags within one stack must have consistent dimensions and packaging.
Boxes	Select the model and template for the actual box. Validate each size and packaging type separately.
Stacking pattern	Regular stacking is required. Supported bag patterns include five-bag interlocking, ring, six-bag and eight-bag arrangements.
Placement	Use a flat, level support surface. Overall bag-stack tilt must not exceed 3°.

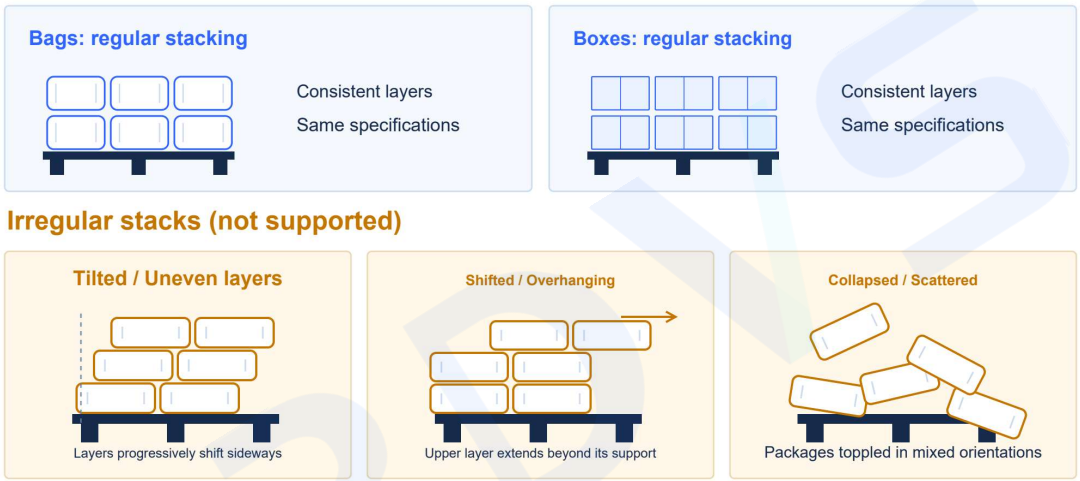


Figure 5. Regular and irregular stacking examples (illustration)

2.2 Limitations

Condition	Limitation and action
Irregular stacks	Tilted or uneven layers, shifted or overhanging packages, and collapsed or scattered stacks are not supported. Restack regularly first.
Severe deformation or damage	Assess severely misshapen or tightly indented bags, and damaged, dented or open boxes individually.
Black, low-reflectance or reflective packaging	Black bags are not supported. Test other low-reflectance or reflective packaging with actual samples to confirm depth quality.
Obstructions or incomplete views	Keep the entire top layer visible. Remove occlusion from grippers, brackets and nearby objects.

Important: After changing the material, packaging or stacking pattern, recheck the model and settings and validate actual picking.

2.3 Operating Environment

Item	Requirement
Lighting	Keep the lights on during operation. Avoid direct sunlight in the work area; shade nearby windows where necessary.
Cleanliness	Prevent oil, dust and water from affecting imaging. Clean the lens regularly.
Multiple cameras	Avoid interference from overlapping fields of view. Check images and point clouds from each camera individually.

2.4 Installation Requirements

Important: Mount the camera level, parallel to the load-supporting surface, with its lens pointing straight down. Secure it firmly and keep the full stack visible from maximum height to the final layer.

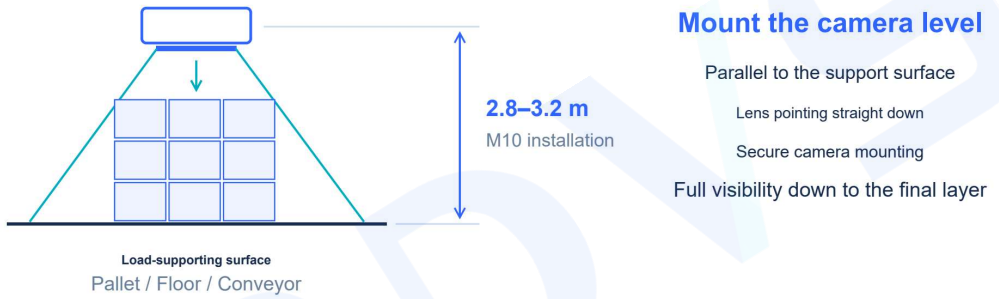


Figure 6. Level installation above a pallet, floor or conveyor

Item	Installation requirement
Camera position	Mount directly above the stack, level and parallel to its support surface, with the lens pointing down. Prevent movement or vibration after installation.
Working distance	For M10, position the camera 2.8–3.2 m above the load-supporting surface. Do not exceed 3.2 m.
Coverage	Keep every layer fully within view. Align the camera's horizontal axis with the long side of the work area.
Compute unit	Install away from dust, water and oil. Leave adequate ventilation space.
Power and cabling	Follow each device manual. Provide the required power supplies and secure power and Ethernet cables.

Important: After moving the camera or changing its mounting position, recheck working distance, work area and hand-eye calibration.

3 First-Time Configuration

3.1 Create and Select a Station

Note: First deploy the Soft Pack Depalletizing camera parameters in AW3. If this is not complete, follow Section 1.1 before continuing.

A station stores the configuration for one working position. Create separate stations for multiple positions and switch between them through communication during automatic operation.

1. Select a camera group on the left, then open Expert > Depalletizing Settings.
2. Select an existing station from the list, or click Add to create one.
3. Keep the station selected. Complete hand-eye calibration, work area setup and material template configuration in that order.

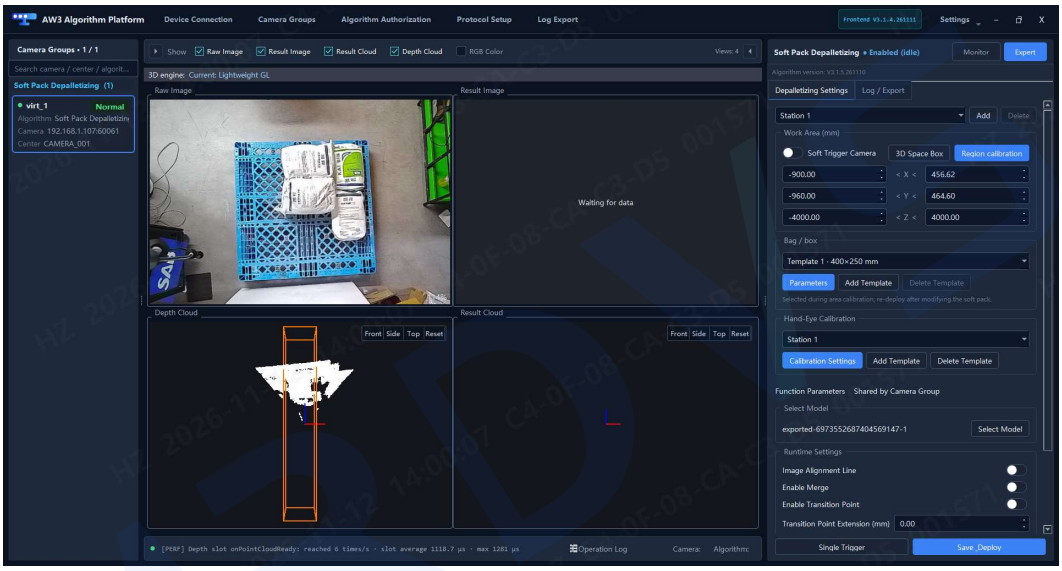


Figure 7. Station selection and Depalletizing Settings

Setting	Scope
Material template	Selected per station; stores package dimensions, filtering and layer settings.
Calibration template	Selected per station; stores hand-eye calibration and tool offsets.
Work Area	Set X, Y and Z detection limits for each station.
Model and Function Parameters	Shared by all stations in the current camera group. Check suitability for other stations before changing them.

Important: Station IDs must match the IDs used in communication commands. Configure and deploy each station separately. Deploy changes before switching stations or leaving the settings page; undeployed changes will be lost.

3.2 Perform Hand-Eye Calibration

Hand-eye calibration maps camera coordinates to robot coordinates. Acquire camera points first, then enter the robot coordinates of the same physical points.

Select the station, then click Calibration Settings under Hand-Eye Calibration.

Important: Fit a calibration probe along the extension of the robot flange Z-axis. Secure the camera and calibration board; neither may move while collecting points.

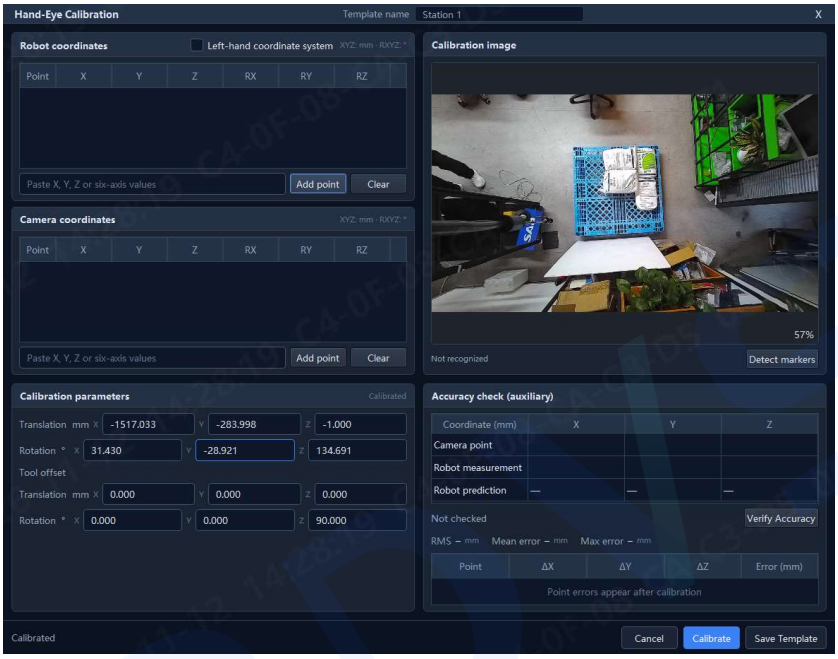
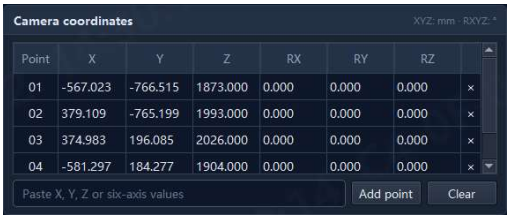


Figure 8. Hand-Eye Calibration

Step 1: Acquire Camera Coordinates

Method 1: Detect Calibration Board Markers (Recommended)

1. Lay the calibration board flat on the stack support plane, centered in the camera view and unobstructed.
2. Click Detect markers. Confirm that four point IDs appear in the image and their coordinates populate Camera coordinates.



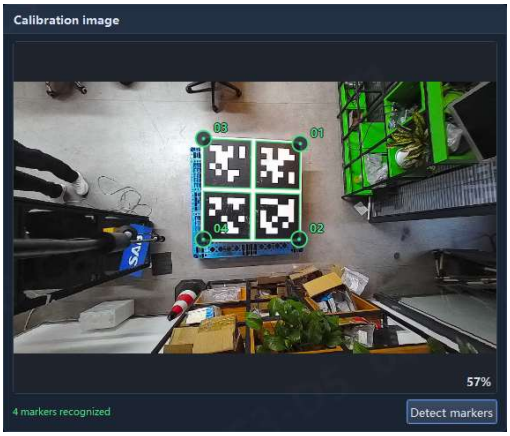


Figure 9. Four detected markers and their camera coordinates

Method 2: Enter Coordinates Manually

1. In Calibration image on the right, right-click the selected calibration point. Click Copy in the coordinate pop-up.

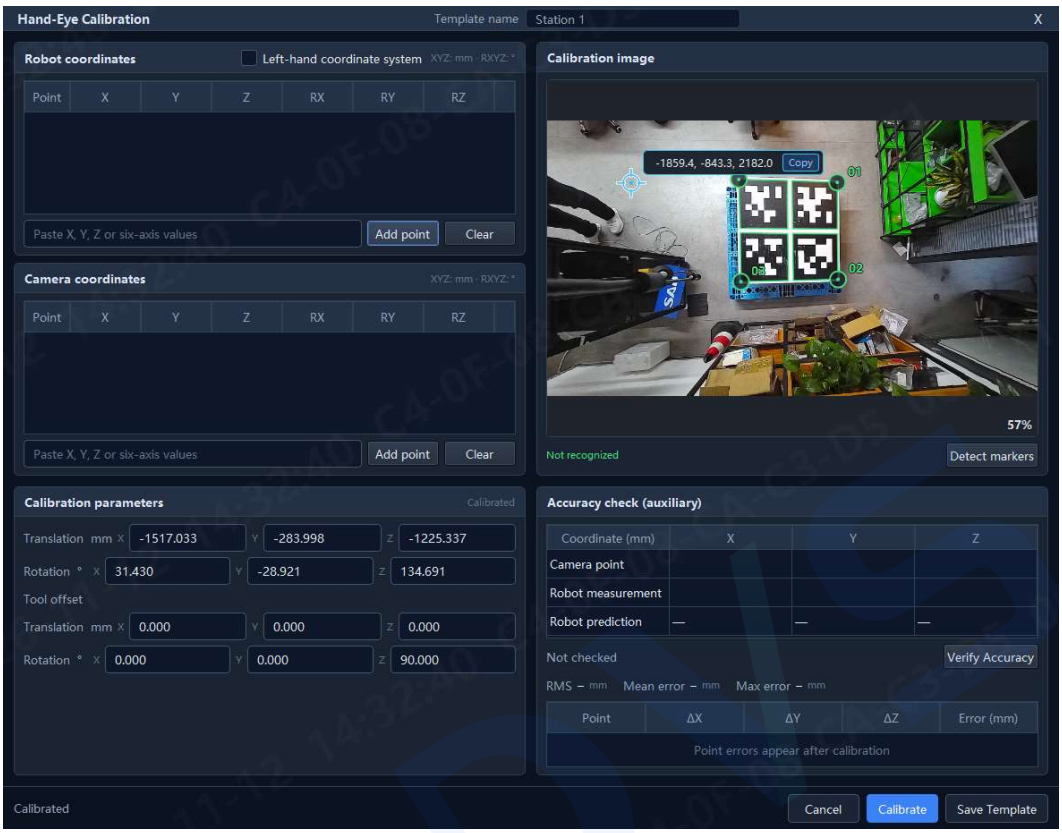


Figure 10. Right-click a point to copy its camera coordinates

2. Paste X,Y,Z, separated by commas, into the Camera coordinates input field. Click Add point.
3. Record four corner points at different positions in sequence. Prefer clear black-and-white intersections and keep track of each physical point.

Note: Choose either method. Manual entry requires 3D camera coordinates, not image pixel coordinates.

Step 2: Enter the Matching Robot Coordinates

1. Follow the camera point order. Move the robot probe tip precisely onto each corresponding feature and read its coordinates from the teach pendant.
2. Enter X,Y,Z, separated by commas, under Robot coordinates and click Add point. Repeat for all four points.

Important: Point count, order and physical locations must match exactly between the two lists. XYZ uses mm. For six-axis input, follow the displayed column order; angles use degrees. Set Left-hand coordinate system to match the robot convention.

Step 3: Calibrate and Review the Results

1. Check both point lists. Delete or reacquire incorrect points, then click Calibrate.

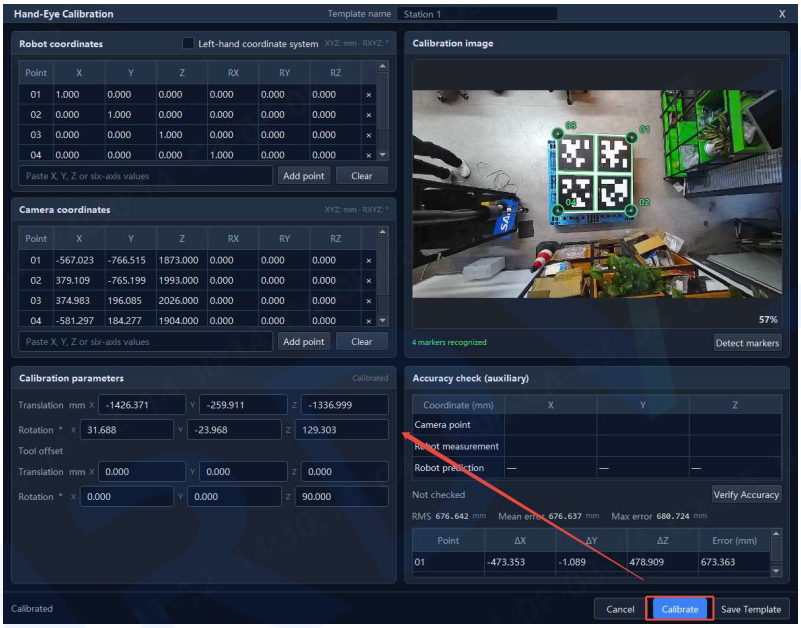


Figure 11. Calibration results showing unacceptable error

Where to look	What to check
Calibration parameters, lower left	Calculated X/Y/Z translation and rotation, and calibration status.
Error statistics, lower right	RMS, Mean error and Max error. The table below lists each point's ΔX, ΔY, ΔZ and total error.
Acceptance	Check the maximum error against the limit, then validate actual picking. Calibrated only means the calculation has completed.

Warning: Maximum calibration error must be below 15 mm and meet project acceptance criteria. The illustrated RMS error is about 677 mm: this calibration is unacceptable and must not be used.

Important: If error is excessive, check point IDs, units, board movement and probe contact positions. Correct the cause, reacquire points and recalibrate.

Step 4: Set the Tool Offset

Compensate for the fixed position and orientation difference between the calibration probe and the actual suction cup or gripper. Enter values in Tool offset under Calibration parameters.

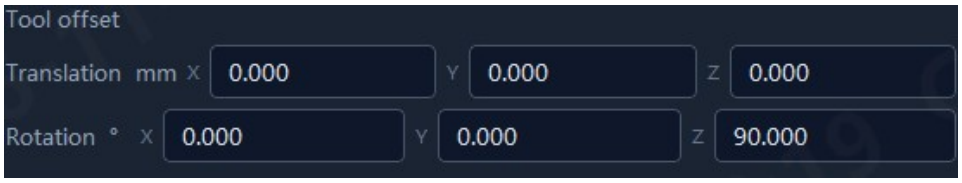


Figure 12. Tool offset fields

Parameter	Entry
Translation X / Y / Z	Fixed positional offset between the calibration probe and actual tool, in mm.
Rotation X / Y / Z	Fixed orientation offset between the calibration probe and actual tool, in degrees.

1. Place one package on the pallet and keep it stationary. Teach the robot to the required pick pose and record it.
2. Move the robot clear. Click Single Trigger and record the vision pose for the same package.
3. Confirm that both poses use consistent frames and tool definitions. Calculate the fixed offset using the project convention and enter it in Tool offset.

Important: Enter translation and rotation separately. Rotation requires coordinate transformation; do not subtract components directly. Z = 90° in the screenshot is an example only.

4. Click Save Template, select it for the current station, then click Save Deploy. Run Single Trigger again to check tool contact position and approach direction.

Position Difference Example

Record the taught and vision positions for the same package in the same coordinate frame:

Record	X (mm)	Y (mm)	Z (mm)
Taught position	100	100	200
Vision position	90	90	240
Difference (taught − vision)	10	10	-40

Note: These values illustrate a position difference only. Transform them into the tool-offset frame before entry. Calculate orientation offsets using the robot's convention; do not subtract Euler angles component by component.

3.3 Set the Work Area

The work area limits detection for the current station. Set each station separately to exclude adjacent stations and surrounding objects.

Step 1: Calibrate the Area Automatically

Important: Automatic region calibration requires the tallest complete, full stack used on site. A partial or shorter stack can produce inaccurate bounds or cause calibration to fail because the region is too small.

1. Select the station and a matching existing material template. If none is available, create one first as described in Section 3.4.
2. Place the tallest complete, full stack in position. Confirm that the entire stack is visible in Raw Image.

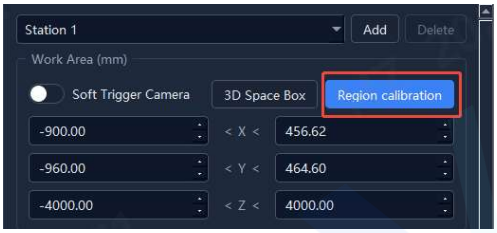


Figure 13. Region calibration control

3. Click Region calibration. On success, the system automatically saves and deploys the settings, then runs one detection.
4. Review the updated X/Y/Z limits, result cloud and recognition results. Confirm that the area covers the stack and excludes nearby objects.
5. If the result is correct, no additional deployment or trigger is needed. If adjustment is required, follow the next step, redeploy and test again.

Possible cause of failure	Action
Stack outside the field of view	Adjust the camera or stack position until the entire target is visible.
Partial stack, insufficient height or region too small	Recalibrate with the tallest complete, full stack. For manual setup, use 3D Space Box.
Image and point cloud do not match	Check streaming and alignment. Restore normal data, then retry.
Unrelated nearby objects included	Use 3D Space Box to reduce or correct the bounds.

Note: If the automatically calibrated region needs adjustment, or manual setup is required, use 3D Space Box as described next.

Step 2: Adjust the Area Manually if Needed

Use 3D Space Box to define the detection volume with minimum and maximum X, Y and Z limits in mm.

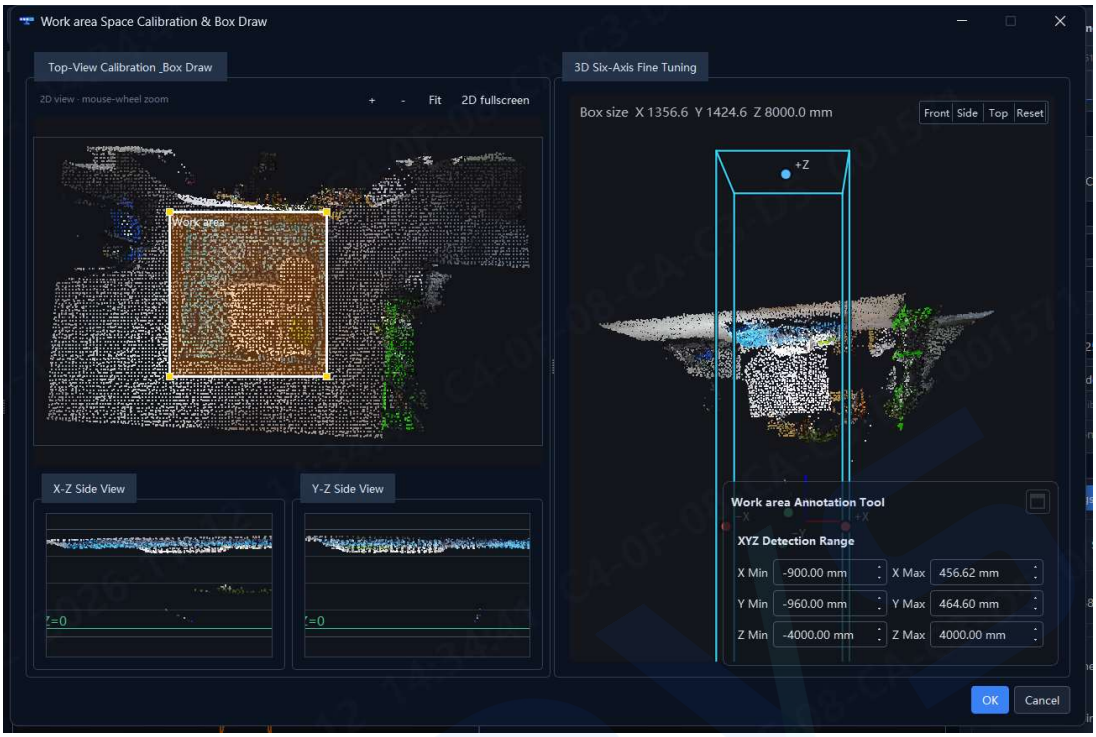


Figure 14. 3D Space Box

1. Under Work Area, click 3D Space Box.
2. Adjust the top-view boundaries to cover the stack and exclude fences, racks and objects on the floor.
3. Check the X-Z, Y-Z and 3D views. Set the height range to cover the full stack through the final layer.
4. Enter precise X/Y/Z limits where needed. Each minimum must be less than its maximum.

Important: Do not confuse work-area coordinates with robot pick coordinates. Use the origin and axes of the work-area frame.

5. Click OK, then Save Deploy to apply the region. Click Single Trigger to verify coverage and recognition.

Important: Recheck the region after changing camera position, mounting angle or stack bounds. Repeat hand-eye calibration if the camera-to-robot relationship changes.

3.4 Configure Material Templates

Material templates store package recognition and picking settings. Create multiple templates for reuse across stations or product changeovers.

1. Confirm the station. Under Bag / box, select a template or click Add Template.
2. Open Parameters, check the Template ID, optionally enter a name and select Package type.
3. Set dimensions, filtering, layer height and sorting as described below.
4. Click Save Soft Pack, then select the saved template for the station. Click Save Deploy to apply it.



Figure 15. Material template parameters

Parameter	Description and setting
Template ID	Unique numeric identifier, assigned automatically at creation. It is fixed after saving and must not be duplicated.
Template Name	Optional. Use a package name or specification for easy selection. The name does not replace the Template ID.
Package type	Choose Bag or Box, or enter a custom English description. See the next page.

- Add: Click Add Template, create a template for a specific type, size or packaging, then save it.
- Reuse: Select an existing template for the station and click Save Deploy. After editing, redeploy it at every station that uses it.
- Delete: Select an unused template and click Delete Template. If a station references it, select another template for that station first.

3.4.1 Select the Package Type

Choose the actual package type. Use a custom English description to emphasize features such as the color of a particular bag.

Type	Use
Bags	For bagged products. Set length, width and thickness from the actual bag.
Boxes	For boxes. Set dimensions and Layer Height from the actual box.
Custom English description	Filter for, or improve recognition of, specific bags using features such as color.

Custom Descriptions

Examples: white bag or blue bag. These are descriptive examples; test recognition performance on site.

Important: Descriptions of objects other than bags and boxes are not recommended. Custom input does not imply support for arbitrary objects.

Test Before Use

1. Enter the description and save the material template. Select it for the station and click Save Deploy.
2. Place the target package in position. Click Single Trigger and check detection boxes, count and poses.
3. Add packages with other colors or features. Trigger again to check filtering and false detections.
4. Repeat with the actual stacking pattern at different heights. Start production only after the results meet requirements.

Check	Pass criteria
Target packages	The required packages are detected consistently.
Other packages	Packages that should be excluded are not selected as targets.
Pick results	Count, order, position and orientation meet site requirements.

3.4.2 Set Dimensions and Layer Height

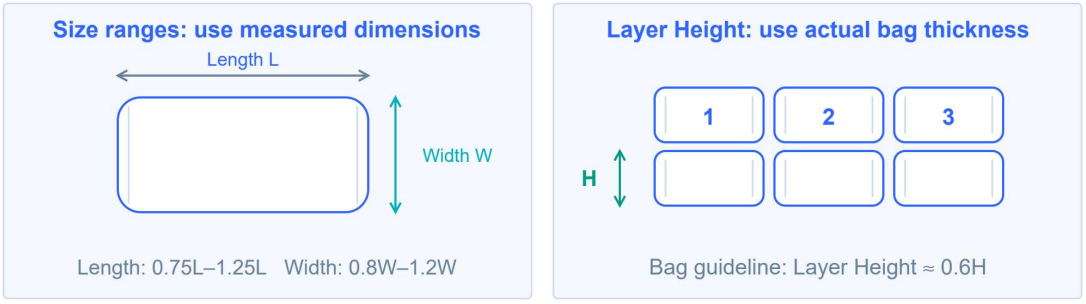


Figure 16. Bag dimensions and layer settings (illustration)

Parameter	Setting guidance
Standard Size (L / W mm)	Enter the measured length and width in mm.
Length Range	Bag guideline: approximately ±25% of actual length. Targets outside the range are filtered out.
Width Range	Bag guideline: approximately ±20% of actual width. Targets outside the range are filtered out.
Layer Height	Separates upper and lower layers. For bags, use approximately 0.6 × bag thickness as a starting point. For boxes, validate against box height and layer spacing.

Example Settings

Measured bag: 600 mm long, 400 mm wide and 200 mm thick. Applying the guidelines above:

Item	Example value
Standard Size	600 × 400 mm
Length Range	450 ~ 750 mm
Width Range	320 ~ 480 mm
Layer Height	120 mm

Note: These are bag-tuning guidelines, not software input limits. Configure boxes using measured dimensions and allowable variation.

3.4.3 Set Filtering and Pick Order

Parameter	Setting guidance
Aspect ratio threshold	Filters unusually proportioned targets. Keep the default initially; first check package type, dimensions and segmentation.
Fill rate	Contour area divided by the area of its minimum rotated bounding rectangle. Recommended: 0.7. Values at or below the threshold are filtered. Check contour segmentation, shape and occlusion first.
Min point count	Minimum number of valid points. Set according to project settings and point-cloud quality. The value 200 shown is an example only.
In-layer Sorting	Sets pick order within a layer; height takes priority by default. Adjust to the robot's picking route and verify.

Tuning Guide

Symptom	First adjustment
Valid packages filtered by size	Check actual length and width against the template. Widen the relevant range slightly if necessary.
Upper and lower layers confused	Check bag thickness or box height. Adjust Layer Height and retest at full, intermediate and final layers.
One package split into several targets	Check creases and model segmentation. Try Enable Merge as described in Section 4.3.
Pick order does not suit the process	Adjust In-layer Sorting. Compare image IDs, Monitor table order and robot motion.

Save and Retest

1. Keep the test scene fixed. Adjust one parameter group at a time and avoid large changes.
2. Save the material template, select it for the station and click Save Deploy.
3. Run Single Trigger. Check count, layer assignment and pick order. Once results are repeatable, validate actual picking.

Note: The interface defines input limits. Where no general value is specified, use the settings supplied for your project.

3.5 Configure Function Parameters

Important: The model and Function Parameters are shared by every station in the camera group. Switching stations does not switch these settings. Check all stations after changes.

1. Click Select Model and choose a model suitable for the package type.
2. Configure merge, transition points and safety detection below. Enable them only where required.
3. Click Save Deploy. Run Single Trigger at each station to verify the new results.

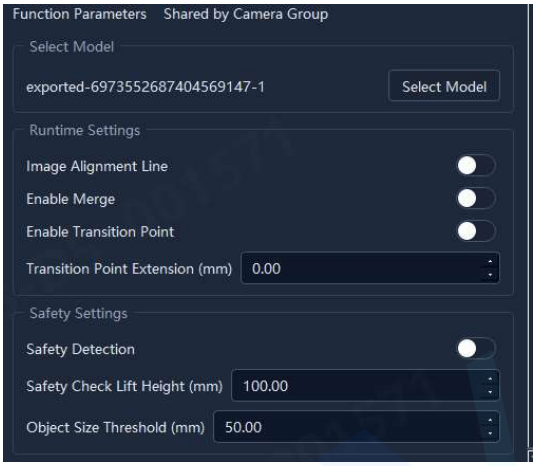


Figure 17. Function Parameters shared by the camera group

Parameter	Purpose and guidance
Enable Merge	Off by default. Enable if one package is split into several targets. Check that neighboring packages are not merged incorrectly.
Enable Transition Point	Enable as needed so the robot first approaches a transition point above the pick point.
Transition Point Extension	Bag guideline: 1.5 × bag thickness. Check the path and increase if contact occurs. Excessive distance can slow the cycle.
Safety Detection	Off by default. Enable to detect objects above the stack, then verify the alarm and control-system response.
Safety Check Lift Height	Height above the current stack top; objects above this level are checked. Bag guideline: 1.5 × bag thickness.
Object Size Threshold	Bag guideline: 30 mm. Objects smaller than this do not trigger an alarm. Consider reducing it if obstructions are missed.
Image Alignment Line	Helps check camera and stack alignment. Display only; changes take effect immediately.

Note: For boxes, validate against box height and the actual pick path. See Chapter 4 for detailed adjustments.

3.6 Deploy Settings and Verify Results

1. Save the material and calibration templates and select them for the station.
2. Click Single Trigger. The system validates, saves and deploys the configuration, then runs one detection after the device confirms deployment.
3. Wait for the new result. Compare the result image and Monitor table for count, layers, pick order and six-axis poses.

Note: Single Trigger verifies settings and recognition. Detection does not run if validation or deployment fails. To apply settings without detection, click Save Deploy.



Figure 18. Targets and pick order in Result Image

Running normally

Package Poses

Station 1 Robot Frame 2 package(s)

Pick Order	X	Y	Z	RX	RY	RZ
1	6.2	-6.2	21.4	39.5	27.5	58.8
2	145.4	233.7	-88.0	31.2	32.4	55.7

Figure 19. Robot poses in Monitor

Check	What to verify
Targets and count	Image targets match the physical packages; the displayed count is correct.
Layers and order	Only currently eligible targets are included. IDs correspond to pick order.
Position and orientation	XYZ and RX/RY/RZ match the site's coordinate and tool conventions.

Warning: Poses shown are examples only. Complete actual calibration and picking tests before using them in operation.

3.6.1 Read the Result Image

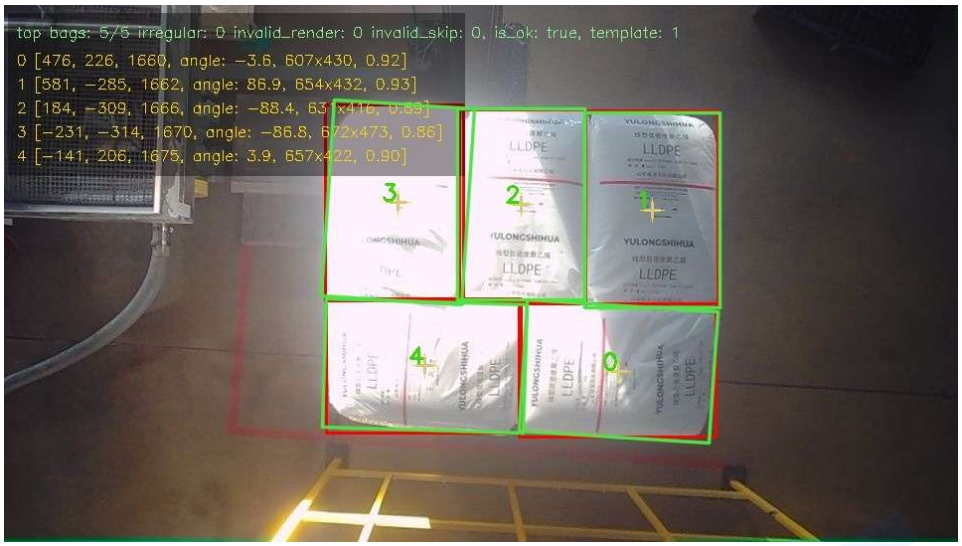


Figure 20. Valid detections and pick IDs

Image marking	Meaning and interpretation
Green box, gold cross and ID	Valid target and its center. Image IDs start at 0; match them to the Monitor table in returned order.
Red detection box	Region detected by the model. The box alone does not indicate an error or final validity.
Blue/orange box or cyan outline	Blue indicates a size or shape rejection. Orange and cyan indicate invalid targets. Read the label for the reason.
W / L / SZ / FR / AR	Size or shape checks. See Section 6.3 for values and actions.
MOUT / OUT / NoPC	Respectively: center samples out of bounds, excessive XY out-of-bounds points within the contour, or no remaining cloud after filtering.
NoPlane	No qualifying point-cloud cluster remains after surface-normal filtering. This target is not output; individual target code 8.
below	Lower-layer target, not picked this cycle. Its size or pose is not necessarily invalid.
M	Depth near the center is insufficient or unreliable, so fallback depth is used. The target may still be valid.
err code / template	Frame-level error code and current template. See Section 6.3 for frame and individual target codes.

Note: Once output packages are confirmed valid and accurately located, no action is needed for other packages' rejection labels; these exclude unsuitable targets. Frame-level errors and obstruction alarms still require attention.

3.6.2 Verify Before Picking

Check coordinates first, then validate actual picking. Meet the criteria below before routine operation.

Item	Use
Camera coordinates	Used for calibration points; match robot points one-to-one by ID.
Robot coordinates	Pick positions and orientations shown in Monitor, obtained through hand-eye calibration and tool-offset transformation.
X, Y, Z	Position in mm.
RX, RY, RZ	Orientation angles in degrees. Axes and angle conventions must match the robot system.
Pick Order	Read in Monitor table order and match each target to the result image.
Tool reference	Check the tool center point (TCP), suction-cup or gripper contact position, and approach direction.

Verification Sequence

1. Align robot base-frame, tool-frame and left-/right-handed conventions.
2. Move the target slightly to verify X, Y and Z directions, one axis at a time.
3. Check calibration error and actual pick error.
4. Confirm targets, layer assignment and pick order before executing motion with the output poses.

Validation item	Pass criteria
Calibration accuracy	Maximum calibration error is below 15 mm and meets project acceptance criteria.
Position and orientation	Tests pass at the center, edges and different heights of the region.
Layer assignment	Only eligible targets are picked. Upper-layer errors do not cause incorrect lower-layer picks.
Result validity	Do not reuse the previous result after no-target detection, obstruction, disconnection or timeout.

Warning: Running normally and Calibrated are software states. Actual picking must still be validated.

3.7 Start Routine Operation

1. Start AW3 and select the camera group. Check images, point clouds and algorithm status.
2. Check the station, material template and calibration template.
3. Run Single Trigger. Check count, pick order and poses.
4. After communication and picking tests pass, start automatic operation under the site's procedure.
5. If a fault occurs, stop using invalid results. Resolve the cause, then trigger again to verify recovery.

Routine Checks

When	Check
After startup	Camera mounting, cables, lens, streaming and algorithm status.
Package specification changes	Switch model or material template; check dimensions and layer settings.
Station changes	Check the region and selected templates. Confirm deployment.
Before picking	Target validity, layer assignment, path and tool condition.
After fault recovery	Trigger again and check the new result. Do not use old poses.

Saving and Applying Changes

Note: Successful automatic region calibration saves and deploys settings and runs one detection. Review that result directly. Later parameter or region changes require deployment and another test.

Action	Effect
Save material template	Updates template parameters. Select and deploy it at the stations that use it.
Save calibration template	Updates calibration and tool offsets. Select and deploy it at the stations that use it.
Save Deploy	Validates, saves and deploys the configuration. Takes effect after device confirmation; does not run detection.
Single Trigger	Validates, saves and deploys automatically, then runs one detection to test settings and recognition.

Important: Click Save Deploy before switching stations or leaving the settings page. Saving a template alone does not deploy the station configuration. Undeployed changes will be lost.

4 Using the Features

4.1 Manage Stations and Templates

Action	Guidance
Add a station	Use for a different work area or pick location. Check its region and templates separately.
Add a material template	Click Add Template. Create and save templates by type, size or packaging for later reuse.
Reuse a template	Select and deploy it at the target station. Redeploy at all affected stations after editing.
Delete a station or template	Select it and click Delete or Delete Template. Reassign any stations referencing a template before deleting it.

Important: Reuse calibration templates only when camera mounting and the robot coordinate relationship are unchanged.

4.2 Manage Models

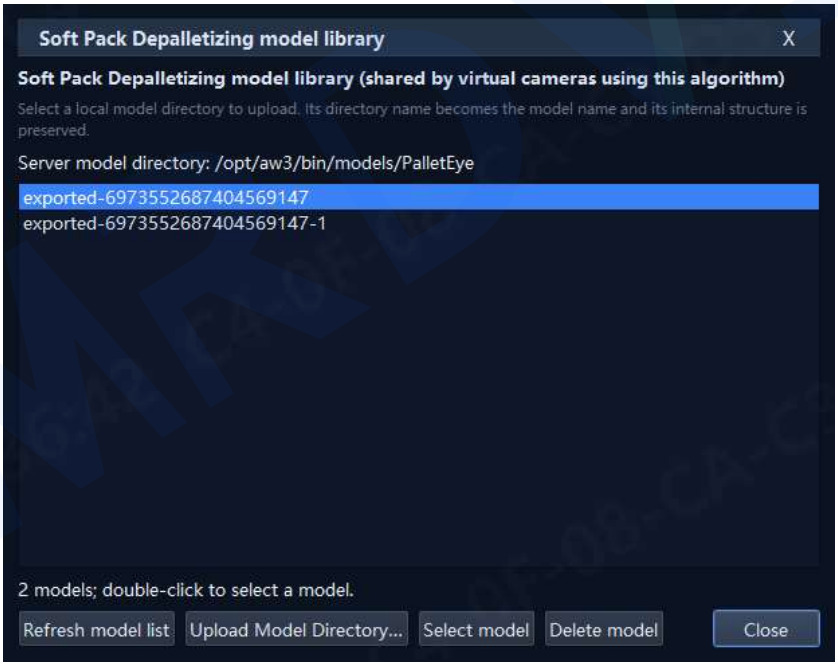


Figure 21. Model library

1. Click Select Model. To add a model, click Upload Model Directory...
2. Upload the complete model directory, click Refresh model list, select the model and click Select model.
3. Check the model name on the main screen. Deploy and trigger a test.

Note: Models recognize targets. Material templates define dimensions and picking rules.

Important: Before deleting a model, check whether other camera groups still use it.

4.3 Merge Detections

Situation	Action and verification
One package detected as several targets	Turn on Enable Merge, deploy and trigger again.
Neighboring packages merged incorrectly	Turn merge off. Check model segmentation and package placement.

4.4 Transition Points

Parameter	Setting guidance
Enable Transition Point	Enable as required by the picking plan. Confirm the receiving controller uses transition poses.
Transition Point Extension	In mm. Check the path and increase if contact occurs. If the cycle is slow, reduce only after validating the path.

Note: For bags, start at approximately 1.5 × bag thickness. For boxes, validate against box height and the actual path.

4.5 Safety Detection



Figure 22. Runtime Settings and Safety Settings

Parameter	Purpose
Safety Detection	Detect obstructions above the pick area. Verify alarms and control-system response after enabling.
Safety Check Lift Height	Sets the vertical position of the detection space, in mm.
Object Size Threshold	Sets the obstruction size threshold, in mm.

- Bag guideline: 30 mm size threshold and lift height of approximately 1.5 × bag thickness.
- If an obstruction alarm occurs, remove the object and confirm the area is clear before resuming.
- Missed alarms: check that detection is enabled and reduce the size threshold if needed. False alarms: increase the threshold or lift height appropriately, then retest.

Note: After changing merge, transition or safety settings, click Save Deploy, then Single Trigger. A successful detection test must be followed by actual picking and control-response tests.

4.6 Image Alignment Guides

Turn on Image Alignment Line to display centerlines and a centering frame in Raw Image for installation checks.

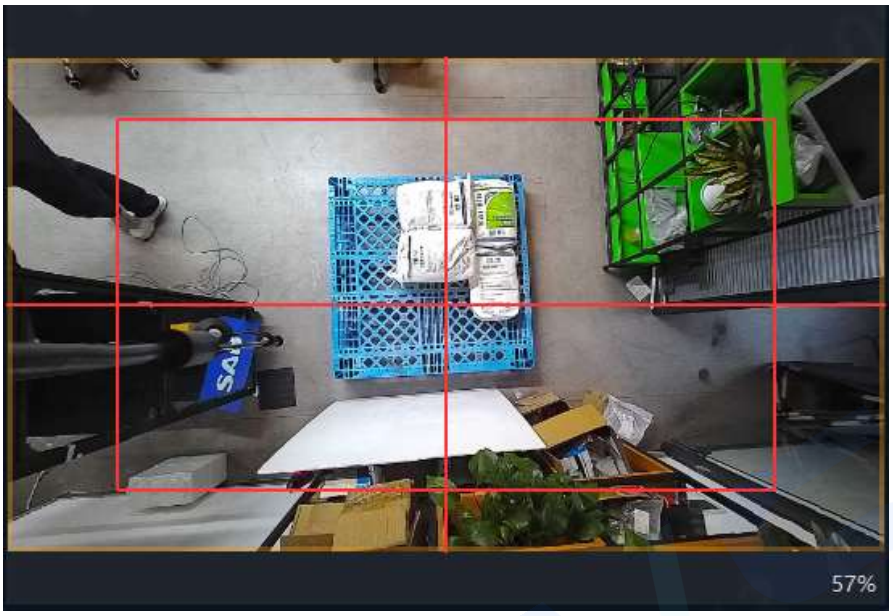


Figure 23. Centerlines and centering frame

Note: Alignment guides do not change the work area or algorithm results.

Important: The image frame is not the 3D detection volume. Work Area settings define that volume.

Checks After Feature Adjustments

Feature	Pass criteria
Merge	One package produces one target; neighboring packages remain separate.
Transition point	Correct extension direction and distance; robot approach path validated on site.
Safety Detection	Obstruction alarms work, and the controller rejects the invalid pick result.
Image Alignment Line	Used to check mounting alignment. Turning it off leaves detection settings unchanged.

Warning: Vision-based obstruction detection does not replace guarding, emergency stops or safety interlocks.

5 Communication and Result Data

5.1 Configure and Test Communication

TCP, UDP, HTTP and Modbus are supported, with customizable message formats. This section describes the JSON result structure and a six-axis output example.

Open Protocol Setup in AW3. Configure connections, trigger commands and result formats for the project.

For detailed communication setup, see Communication Protocol Configuration in the AW3 User Manual.

1. Select the protocol. Check connection roles, addresses, ports and receiving-system requirements.
2. Configure output fields. Use consistent coordinate frames, position units and angle conventions.
3. In Depalletizing Settings, click Save Deploy, then Single Trigger. Compare the interface with received messages and test both normal and error handling.

5.2 Read the Result Fields

Field	Meaning and interpretation
ret_code	Return code; 0 indicates success in this example. See Section 6.3 for errors.
error_str	Result or error description. In this example, 成功 means success.
grasping_safe	Pick-safety flag. If false, do not use this result to execute a pick.
template_name	Current template identifier as a string; "1" in this example.
results	Array of target results. Use its length as the count; an empty array means no target results.
result_index	Zero-based target index. Match image IDs and read results in returned order.
x, y, z	Position extracted from the pose matrix translation, in mm. Retains the original matrix's coordinate frame.
rx, ry, rz	Orientation angles converted from the rotation component, in degrees. This example uses the ZYX convention.
tl_2d / br_2d	Top-left / bottom-right detection-box coordinates, [u, v], in pixels.
length_3d / width_3d	Measured target length / width, in mm.

Note: Check the return code, safety flag and result array before reading poses. Do not reuse previous-frame coordinates when targets are absent, an error occurs or no new result is available.

Six-axis conversion changes pose representation only; it does not transform camera coordinates into robot coordinates. Verify hand-eye calibration, tool offsets and the output frame before use.

5.3 JSON Six-Axis Output Example

The status, template and target fields are retained from the current message. rot_matrix is converted to x, y, z, rx, ry, rz. The example contains four targets from one frame.


```
{
  "error_str": "成功",
  "grasping_safe": true,
  "results": [
    {
      "result_index": 0,
      "x": -1332.67, "y": 1160.681, "z": 916.154,
      "rx": 2.045, "ry": -9.913, "rz": 2.754,
      "tl_2d": [319.0, 308.0], "br_2d": [411.0, 445.0],
      "length_3d": 798, "width_3d": 545
    },
    {
      "result_index": 1,
      "x": -283.92, "y": 562.169, "z": 979.139,
      "rx": 8.008, "ry": 4.108, "rz": -88.78,
      "tl_2d": [462.0, 224.0], "br_2d": [603.0, 317.0],
      "length_3d": 843, "width_3d": 532
    },
    {
      "result_index": 2,
      "x": -716.686, "y": 1187.421, "z": 987.653,
      "rx": 3.076, "ry": -3.802, "rz": 4.999,
      "tl_2d": [419.0, 311.0], "br_2d": [507.0, 442.0],
      "length_3d": 783, "width_3d": 528
    },
    {
      "result_index": 3,
      "x": -200.294, "y": 1230.174, "z": 1015.417,
      "rx": 2.481, "ry": -15.118, "rz": 0.515,
      "tl_2d": [507.0, 312.0], "br_2d": [590.0, 445.0],
      "length_3d": 811, "width_3d": 503
    }
  ],
  "ret_code": 0,
  "template_name": "1"
}
```


These six-axis fields are an example output mapping and must be configured in the custom protocol. If the robot uses another orientation convention, convert accordingly; renaming fields alone is insufficient.

6 Troubleshooting and Maintenance

6.1 Device and Streaming Issues

Symptom	Checks and actions
No RGB image or point cloud	Check power, Ethernet cables, IP addresses, camera group and connection status.
Image stops updating	Restore streaming, then trigger again. Do not use results from an old image.
Authorization or algorithm status error	Check software version and authorization as prompted. See the AW3 User Manual.
Streaming fails after using LanxincameraViewer	In Device Connection / Base Camera, open Central Service and select Restart current center. No restart is needed when streaming is normal.
Results visible in AW3 but no external data	Check protocol enablement, camera-group association, address and port.

6.2 Recognition and Picking Issues

Symptom	Check first	Verify after correction
Valid packages rejected	Type, size ranges, depth cloud and region.	Valid targets retained; unrelated objects excluded.
One package split into multiple detections	Model, creases and Enable Merge.	One package corresponds to one target.
Incorrect layer assignment	Layer Height and region Z limits.	Top and lower layers correctly distinguished.
Persistent pick offset	Calibration point pairs, frames, TCP and tool offsets.	Acceptable picks at multiple positions and orientations.
Missed or false obstruction alarms	Missed: enable detection or reduce the size threshold. False: increase the threshold or lift height appropriately.	Correct alarm and control-system response.
Incorrect results after switching stations	Selected templates, region and deployment status.	Settings match the active station.

Note: Restore images and point clouds before tuning algorithm settings. Change one parameter group at a time, deploy and retest.

6.2.1 Size and Region Errors

Size Mismatch



Figure 24. Size mismatch: W/L measurement labels

Symptom: W/L indicates that width or length failed the template check. W424 in the image means a measured width of 424 mm. A red detection box alone is not an error.

Action: See Section 3.4, Configure Material Templates. Check Standard Size, Length Range and Width Range. Click Save Deploy after changes, then test with Single Trigger.

Outside the Work Area

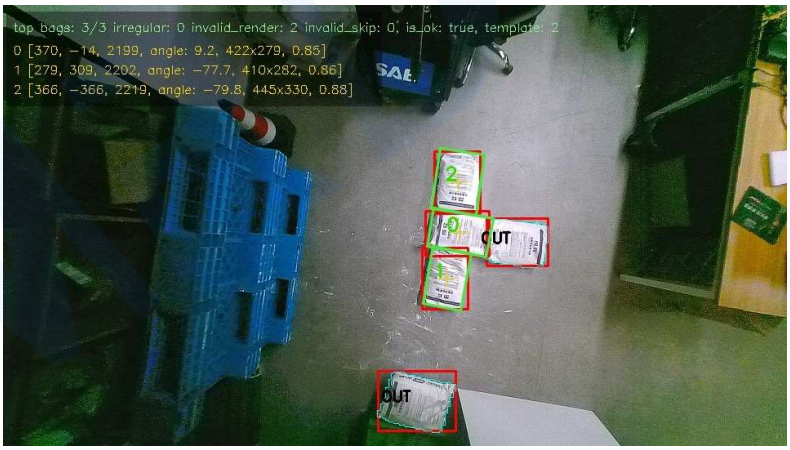


Figure 25. Out-of-area target: OUT label

Symptom: OUT means over 5% of sampled contour pixels have corresponding 3D points outside the XY bounds. This target is not output.

Action: Move the target inside the work area, recalibrate the region or adjust it manually. Exclude unrelated objects, then trigger again.

6.2.2 No Targets and Obstruction Alarms

No Valid Targets

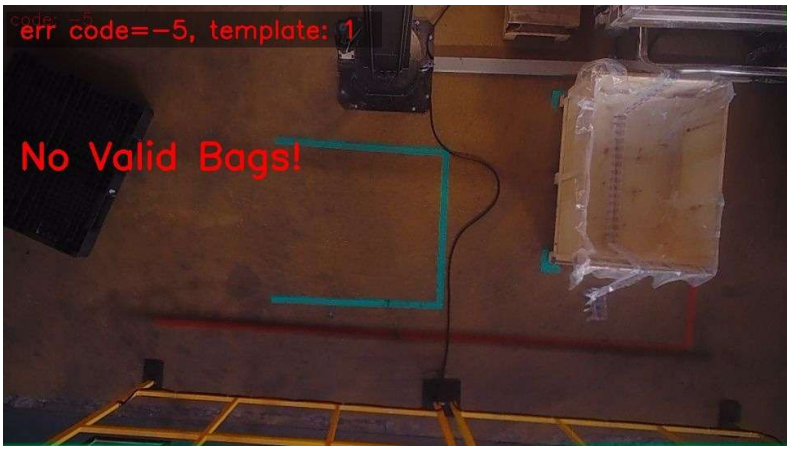


Figure 26. No valid targets: No Valid Bags!

Symptom: No valid targets remain. The image displays No Valid Bags! and error code −5.

Action: Confirm packages are present and the region is correct. If packages remain, check type, dimensions and point-cloud filters.

Obstruction Alarm



Figure 27. Obstruction alarm: OBSTACLE DETECTED!

Note: The yellow box and arm label were added manually for explanation; they are not software detection output.

Symptom: The image turns red and displays an obstruction alarm. Do not use this result for picking.

Action: Remove objects above the pick area and have the operator confirm clearance before triggering again. For false alarms, check Object Size Threshold and Safety Check Lift Height.

6.3 Error Codes and Actions

Note: Once output packages are confirmed valid and accurately located, no action is needed for other packages' rejection labels; these exclude unsuitable targets. Frame-level errors and obstruction alarms still require attention.

Frame-Level Errors

Code	Result image label	Meaning	Action
0	Target statistics at top	Detection successful.	No action required.
−5	No Valid Bags! code: -5 / err code=-5	No valid packages output.	Confirm packages are present. Check target labels for rejection reasons.
−6	OBSTACLE DETECTED!	Obstruction above the pick area.	Remove arms, grippers or other objects. For false alarms, check safety settings (4.5).
−7	UPPER FILTERED - BLOCK LOWER PICK	Upper-layer package rejected; lower-layer picking blocked.	Check labels on top-layer packages. If necessary, manually remove the problem package and retry.
−9	BAG OUT OF IMAGE!	Target failed the image-edge check.	Reposition the stack fully within view and clear of image edges. Recalibrate if the camera moves.
−10	—	Input image empty or wrong size.	Check Raw Image and device connections (1.1). Restore capture and retry.
−101	—	Algorithm settings not configured or deployed.	Complete station configuration and click Save Deploy (3.6).

Note: “—” means no fixed image label has been confirmed. Use the return code.

Note: For unlisted codes, or if no valid output remains after troubleshooting, export data as described in Section 6.4 and contact technical support.

Individual Target Filtering

Note: Investigate only when required packages are missing or results are inaccurate. Numbers in labels are examples.

Code	Result image label	Meaning	Action
1	NoDepth	No valid depth near the target center.	Check depth gaps, occlusion, reflections and mounting distance (Chapter 2).
3	FR 0.61	Contour fill rate does not exceed the threshold.	Check segmentation and occlusion. Recommended Fill rate: 0.7 (3.4.3).
5	OUT	Too many target points outside the XY region.	Move the package into the region or correct its XY bounds (3.3).
7	NoPC	No cloud remains after depth and region filtering.	Check raw depth, then the region's X/Y/Z limits (3.3).
8	NoPlane	No valid surface point-cloud cluster remains after surface-normal filtering.	Check camera leveling, package tilt, surface cloud and Min point count (3.4.3).
9	MultiClu	Multiple clouds within one target triggered filtering.	Eliminate airborne dust and occlusion. Check cross-layer segmentation and Layer Height (3.4.2).
10	W620	Width mismatch; number is measured width in mm.	Check package width and template Width Range (3.4.2).
11	L920	Length mismatch; number is measured length in mm.	Check package length and template Length Range (3.4.2).
12	SZ 920x620 or W420,L300	Length/width combination failed; not necessarily both out of range.	Confirm the contour covers one complete package. Check template size and ranges (3.4.2).
13	—	Poor point-cloud quality.	Reduce airborne dust and retry once the view clears. If unresolved, export data (6.4).
14	below	Lower-layer package; not output this cycle.	No action for correct layer filtering. If misclassified, check Layer Height (3.4.2).
19	AR 4.82	Image contour aspect ratio meets or exceeds the limit.	Check for neighboring packages segmented as one. Check Aspect ratio threshold (3.4.3).
20	MOUT	3D samples near the center are out of bounds.	Check center depth first, then region X/Y/Z limits (3.3).

Important: Set parameters for the actual packages. After changes, click Save Deploy and verify with Single Trigger.

Other Error Codes

Note: Use this supplementary table only when output is missing or inaccurate. If the checks do not resolve the issue, export images, point clouds and logs as described in Section 6.4 and contact technical support.

Code	Result image label	Meaning	Action
−1	—	Insufficient valid cloud in the work area.	Confirm packages are present. Check depth and region limits.
−2	—	Model failed or returned no result.	Select and deploy the model, then trigger again.
−3	—	Insufficient cloud after filtering.	Check missing depth, region limits and Min point count.
−4	—	No usable point-cloud cluster.	Remove occlusion. Check region limits and Min point count.
−100	—	Unknown error.	Save site data and contact technical support.
2	E2	No usable package contour.	Remove occlusion. Check Package type and model.
4	E4	Image contour too small.	Ensure the package is fully visible. Check mounting distance.
6	E6	Insufficient contour points.	Remove occlusion and check contour completeness.
16	BadCtr	Invalid 3D center.	Check center depth and region Z limits.
17	E17	Overlapping candidate filtered out.	No action if correct output exists. Otherwise check segmentation and merge settings.
21	E21	No valid orientation obtained.	Check surface cloud completeness and excessive package tilt.

Note: After parameter changes, click Save Deploy, then verify with Single Trigger.

6.4 Export Diagnostic Data

1. Record the fault time, camera group, station and selected templates.
2. Save Raw Image, Result Image, point clouds and error messages.
3. Export logs and data from before and after the fault. Include reproduction steps and recent setting changes.

For log and data export instructions, see Log Export in the AW3 User Manual.

Data	Include
Versions and devices	AW3 and algorithm versions, camera model and connection details.
Site and configuration	Package photos, dimensions, stacking pattern, station, model and template settings.
Fault record	Time, result image, point clouds, codes and logs.
Reproduction details	Trigger conditions, steps, frequency and recent changes.

6.5 Maintenance and Recalibration

Change or symptom	Recheck
Camera position, height or angle changes	Field of view, region, hand-eye calibration and actual picking.
Robot base frame or mounting relationship changes	Coordinate conventions, hand-eye calibration and tool offsets.
Suction cup or gripper replaced	TCP, tool offsets, approach direction and path.
Package dimensions, packaging or stacking pattern changes	Model, material template, layer and safety settings.
Stack position or bounds change	Work-area coverage and boundaries.
Model or software upgrade	Parameter compatibility, normal/error results and communication.

- Clean the lens regularly. Check cable security and device ventilation.
- Record active settings before changes. Afterwards, click Save Deploy and Single Trigger. Resume automatic operation only after verification passes.

Revision History and Technical Support

Version	Date	Changes
v1.0	2026-11-10	Initial release. Covers bags and boxes: configuration, calibration, features, communication and troubleshooting.

Technical Support

Zhejiang MRDVS Technology Co., Ltd.

Item	Contact information
Hotline	+86 400-025-6680
Website	www.mrdvs.com
Knowledge Community	https://docs.mrdvs.com/en
GitHub	https://github.com/Lanxin-MRDVS
LinkedIn	https://www.linkedin.com/company/mrdvs

Figure 28. Visit MRDVS on GitHub